

grblHAL Teensy 4.x — Fork Changes

Changes carried on `stevenwood`'s grblHAL forks for the iMXRT1062 (Teensy 4.1) build. Each card was originally proposed upstream; as of **2026-06-18 those PRs are withdrawn**, and every change is now staged as a same-repo PR on the fork (base = the fork's own master) for review and cherry-pick.

How to consume these changes

- **Use the whole set as-is** — `git clone -b srw/local-build-config https://github.com/stevenwood/iMXRT1062`, run `./srw/setup_forks.sh`, overlay `srw/combined` per `srw/README.md`, and build.
- **Cherry-pick one** — each change is a single-purpose branch (see its card and `firmware-forks.json`); `git cherry-pick` it onto your own fork.
- **Submit one upstream** — open a PR to the matching grblHAL/* repo from the branch (or reopen the withdrawn PR).

Reviewable fork PRs: core [#1](#) · [#2](#) · [#3](#) · [#4](#) | Plugin_SD_card [#1](#) · [#2](#) | Plugin_networking [#1](#)

1. Fix grblHAL hang when soft-reset arrives during \$F= SD streaming

Withdrawn — fork PR #1

Repo: `grblHAL/Plugin_SD_card` · Branch: `stevenwood:fix/reset-during-sd-streaming` · Files: `fs_stream.c`

The cleanup branch in `onReset()` required `hal.stream.type == StreamType_File`, which transiently becomes false during tool-change `suspend_read` and `on_stream_changed` callbacks. A reset arriving in that window leaks the file handle and the stream redirection. Symptom: grblHAL hangs and `ioSender` access-violates on the next command; only a Teensy power-cycle recovers it.

Verified locally: reproducible pre-fix (reset during install-probe M0 hold), clean recovery post-fix. Tightened guard fires when EITHER `file.handle` is open OR `active_stream.type` is non-Null; `stream_end_job` is already idempotent.

PR #13

View diff

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2. Add DEFAULT_MACRO_ATC_OPTIONS build-time default for \$675

Withdrawn — fork PR #2

Repo: `grblHAL/Plugin_SD_card` · Branch: `stevenwood:feat/default-macro-atc-options` · Files: `macros.c`

The `$675` setting is currently restored to `0` hardcoded on first boot or `$RST=$`. A build-time `#define DEFAULT_MACRO_ATC_OPTIONS <n>` lets drivers ship firmware where the setting defaults to something else — useful when senders want to detect ATC capability via the `ATC=0/ATC=1` token in `$I` rather than the generic `TC`, fallback.

Follows the same `#ifndef DEFAULT_XXX` pattern used throughout `config.h` for other settings. Default behavior unchanged.

[PR #14](#)[View diff](#)[Browse branch](#)

3. Honor `LITTLEFS_ENABLE=2` as `YMODEM` request even when `SDCARD` is enabled

Withdrawn — fork PR #1

Repo: `grblHAL/core` · Branch: `stevenrwood:feat/littlefs-y modem-combined` · Files: `driver_opts.h`

Pre-existing code in `driver_opts.h` demotes `LITTLEFS_ENABLE` from 2 to 1 whenever `SDCARD` is also enabled (to handle the root-mount conflict). The demotion runs before `FS_ENABLE` is computed, so the `"=2` implies `YMODEM`" semantic — which works for LittleFS-alone — was unreachable in the combined config. `NEWOPT` advertised `FS,SD` instead of `YM,SD`; `y modem_init()` never fired.

Captures the user's `=2` intent into a new `LITTLEFS_YMODEM_REQUESTED` flag before demotion. `FS_ENABLE` in the combined case grants `YMODEM` if either `SDCARD_ENABLE==2` OR `LITTLEFS_YMODEM_REQUESTED`. Mount-dir logic tightened to only put LittleFS at `/` when `SDCARD` is absent at build time. Tested end-to-end on Teensy 4.1.

[PR #966](#)[View diff](#)[Browse branch](#)

4. Add `[BUILD:...] line to $I` from generated `build_stamp.h`

Withdrawn — fork PR #2

Repo: `grblHAL/core` · Branch: `stevenrwood:feat/build-timestamp-line` · Files: `report.c`

`GRBL_BUILD` in `grbl.h` is a hand-edited integer literal that only changes when an upstream release is tagged. Operators rebuilding firmware locally have no way to confirm via `$I` that the running `.hex` matches the one they just produced. This PR adds a `[BUILD:<date> <time>]` line guarded by `__has_include("build_stamp.h")` — drivers that ship a `build_stamp.h` with a `BUILD_TIMESTAMP` macro get the line; drivers that don't are unaffected.

Marked Discuss because Terje may prefer his own build-stamping approach. Companion driver-side plumbing (a PlatformIO pre-build script regenerating `build_stamp.h` each build) lives in each driver,

not core — keeps this PR minimal.

[PR #967](#)[View diff](#)[Browse branch](#)

5. Report mDNS hostname in \$I boot info

Withdrawn — fork PR #1

Repo: [grblHAL/Plugin_networking](#) · Branch: [stevenrwood:feat/hostname-boot-info](#) · Files: [wiznet/enet.c](#)

\$I currently emits [IP:...] but not the hostname being broadcast over mDNS. Diagnosing "grblhal.local won't resolve" is hard when you don't know whether mDNS is silently failing, broadcasting a different name than you expect, or running fine on a network where the client just can't see it. This PR emits [HOSTNAME:<name>.local] immediately after [IP:...] when MDNS_ENABLE is set.

Single-line addition guarded by MDNS_ENABLE and a non-empty hostname check. .local suffix appended in the output (not stored) so the operator sees the exact form clients should query.

[PR #22](#)[View diff](#)[Browse branch](#)

Driver-local patches — kept in the driver fork, not submitted upstream

SRW build config for BOARD_T41U5XBB on Teensy 4.1

Driver-local

Repo: [stevenrwood/iMXRT1062](#) (driver fork) · Branch: [srw/local-build-config](#) · Files: [grblHAL_Teensy4/src/my_machine.h](#), [grblHAL_Teensy4/platformio.ini](#), [grblHAL_Teensy4/scripts/touch_build_stamp.py](#), [.gitignore](#)

Carries the SRW build-time config so any machine that clones [stevenrwood/iMXRT1062](#) at this branch builds the same firmware. [my_machine.h](#) mirrors the Web Builder JSON (17 symbols) plus two unmerged-PR features (`LITTLEFS_ENABLE=2` for combined SDCARD+LittleFS YMODEM via PR #966, and `DEFAULT_MACRO_ATC_OPTIONS=2` via PR #14). [platformio.ini](#) wires in a pre-build script that regenerates [build_stamp.h](#) with the current datetime each compile, so the [BUILD:...] line from PR #967 reflects the actual build moment.

Lives on its own branch so it doesn't interfere with future master updates from upstream and isn't confused with the PR branches. To use on a fresh checkout: `git clone -b srw/local-build-config https://github.com/stevenrwood/iMXRT1062.git`, then `git submodule update --init --recursive`, then run `setup_forks.sh` from this repo to add the fork remotes and pull the 5 PR branches into the relevant submodules.

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Case-insensitive LittleFS name matching Driver-local

Repo: [stevenrwood/IMXRT1062](#) (driver fork) · Branch: [pr/littlefs-case-insensitive](#) — now **folded into** [srw/local-build-config](#) · Files: [grblHAL_Teensy4/src/littlefs/lfs.c](#), [grblHAL_Teensy4/platformio.ini](#)

The g-code parser upper-cases O-word labels, so `O<cal> CALL` looks for `/littlefs/CAL.macro` — but LittleFS is case-sensitive while FatFs (SD) is case-insensitive, and the VFS treats the two filesystems interchangeably. A lower-case `cal.macro` on LittleFS then isn't found, and (because `ngc_flowctrl` masks the file-open failure) it surfaces misleadingly as `error:81 Unknown flow statement`. A build flag `LFS_CASE_INSENSITIVE` adds case-folded name comparison so LittleFS matches FatFs semantics — making the upper-casing a non-issue for *every* file operation, not just macro calls.

Not an upstream PR: it patches the driver's **vendored** `littlefs/lfs.c`, which Terje is unlikely to accept as a core/plugin change. Kept driver-local in the `stevenrwood/IMXRT1062` fork instead. (Considered and rejected: a core-side `CALL` fallback that lower-cases the macro filename — narrower, only covers the macro path, and doesn't make the two filesystems behave consistently.)

[View diff](#)[Browse branch](#)

Flow-control whitespace tolerance (`O<cal> CALL`) Driver-local — fork PR #4

Repo: [stevenrwood/core](#) · Branch: [fix/ngc-flowctrl-whitespace](#) (also in the [srw/combined](#) overlay) · Files: [ngc_flowctrl.c](#)

`grblHAL` keeps interior spaces in a block (`gc_normalize_block` strips only leading whitespace), so `O<cal> CALL` reaches `read_command` with the gap intact and the keyword `read` lands on the space → `error:81 Unknown flow statement`. Skip leading spaces/tabs before reading the flow keyword so an O-word call parses with or without the gap — hand-typed at the MDI, sent by a sender that doesn't close it up, or read from a file.

Driver-local core change, not previously filed upstream. Small and defensible; could be upstreamed later (open a PR to `grblHAL/core` from this branch).

[Fork PR #4](#)[View diff](#)[Browse branch](#)

O<name> CALL macro search path for `LITTLEFS_ENABLE=2` Driver-local — fork PR #3

Repo: [stevenrwood/core](#) · Branch: [fix/littlefs-macro-search-path](#) (also in the [srw/combined](#) overlay) · Files: [ngc_flowctrl.c](#)

The named-sub CALL handler searched `/littlefs` only under `#if LITTLEFS_ENABLE == 1`. With `LITTLEFS_ENABLE=2` (YModem) and an SD card present (LittleFS at `/littlefs`, root = SD), it searched root only, so `0<cal>` CALL never found `/littlefs/cal.macro` — the open failure was then masked as `error:81`. Gate the `/littlefs-then-root` search on `#if LITTLEFS_ENABLE (1 or 2)`. This was the actual root cause of the long `0<cal>` CALL saga (whitespace + case were red herrings on top of it).

Driver-local core change, not previously filed upstream — a genuine bug in the `LITTLEFS_ENABLE=2` path from the `littlefs-y modem` work (fork PR #1). Could be folded into that branch or upstreamed on its own.

[Fork PR #3](#)[View diff](#)[Browse branch](#)

Add PRs from Windows laptop: when you push additional branches from the laptop, append a new `<div class="pr">` block here using the same template — repo, branch, summary, why, actions. Two URL patterns are useful:

Compare page (before the PR is filed): <https://github.com/grblHAL/<REPO>/compare/master...stevenrwood:<REPO>:<BRANCH>?expand=1>

Live PR (after `gh pr create`): <https://github.com/grblHAL/<REPO>/pull/<N>>

Forks created: [core](#) · [Plugin SD card](#) · [Plugin networking](#) · [iMXRT1062](#) (driver, for driver-local patches)